Optimum and Adaptive Signal Processing Systems



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- Random Variables and Signals
  - Random Variables
  - Random Vectors, Multivariate Gaussian Distribution
  - Random Signals (Stochastic Processes)
- Basics of Estimation Theory with Applications in Signal Processing
  - Classical Estimation: CRLB, MVU, BLUE, ML, LS, Estimation of ACS and PSD of WSS processes
  - Bayesian Estimation: MMSE, MAP, LMMSE
- Optimum Filters: Wiener and Least Squares Filters
  - Wiener Filters
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  - Applications: (Inverse) System Identification, Linear Prediction (e.g. for speech), Noise Cancellation
- Adaptive Filters (and Applications)
  - LMS
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- Kalman Filter and Extended Kalman Filter (with Applications)

Content
Classical Estimation

- Basics
- Unbiased Estimators, Minimum Variance Unbiased Estimators (MVU)
- Cramer Rao Lower Bound
- Linear Models
- Best Linear Unbiased Estimator (BLUE)
- Maximum Likelihood Estimation
- Least Squares
- Estimation of ACS and PSD of WSS processes

Estimation Theory – Classical Estimation Basics

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- Modern estimation theory can be found at the heart of many electronic signal processing systems designed to extract information.
- These systems include radar, sonar, speech, image analysis, biomedicine, communications, control,… and many more.
- In all these systems we are faced with the problem of extracting values of parameters based on continuous-time waveforms.
- Due to the use of digital computers to sample and store the continuous-time waveform, we have the equivalent problem of extracting parameter values from a *discrete-time* waveform or a *data* set.

Estimation Theory – Classical Estimation Basics

- Mathematically, we have the *N*-point data set

  $\{x[0], x[1], \ldots, x[N-1]\}$

  which depends on an unknown parameter θ.

- We wish to determine θ based on the data or to define an *estimator*

$$\hat{\theta} = g(x[0], x[1], \ldots, x[N-1])$$

  where *g* is some function.

- This is the problem of *parameter estimation.*

Estimation Theory – Classical Estimation Basics

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- An *estimator* $\hat{\theta} = \mathrm{g}(\mathbf{x})$ may be thought of as a *rule* that assigns a value to θ for each realization of

$$\mathbf{x} = \left[x[0], x[1], \ldots, x[N-1]\right]^T.$$

  An estimator is a random variable. As such its performance can only be completely be described by its PDF.

- The *estimate* of θ is the *value of* $\hat{\theta}$ obtained for a *given* realization of **x**.
- This distinction is analogous to a random variable and its realization.
- Although some authors distinguish between the two by using capital and lowercase letters, we will not do so. From here on we also do no longer distinguish between a RV and its realization.
- The meaning will, hopefully, be clear from the context.

Estimation Theory – Classical Estimation Basics

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- In determining good estimators, the first step is to mathematically model the data.

- Because the data are inherently random, in classical estimation we describe it by its joint probability density function

$$p(x[0], x[1], \ldots, x[N-1]; \theta) = p(\mathbf{x}; \theta),$$

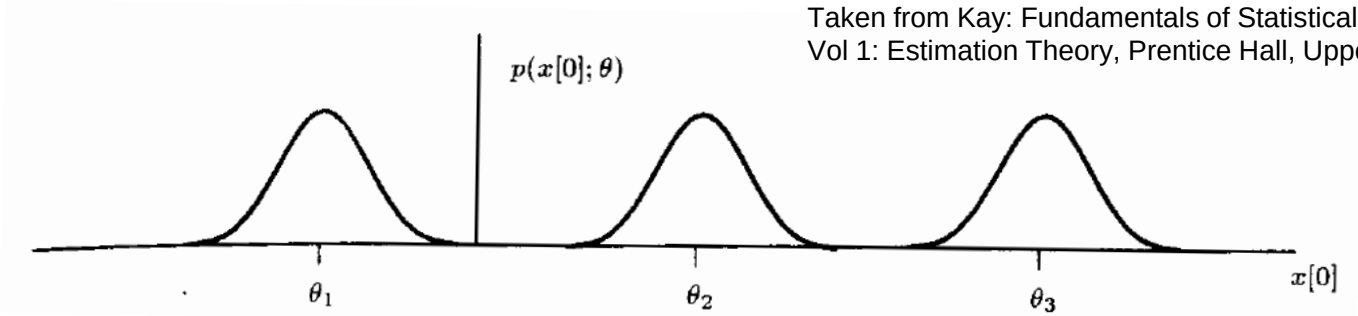
  which is ***parameterized*** by the unknown parameter θ, i.e., we have a class of PDFs, where each one is different due to a different value of θ. We will use a semicolon to denote this dependence.

- Example: Assume we make one noisy measurement of a DC-voltage: $x[0] = \theta + w$, then the PDF of the data might be

$$p(x[0]; \theta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(x[0] - \theta)^2\right).$$

Estimation Theory – Classical Estimation Basics

- Three examples of the class of PDFs:



Taken from Kay: Fundamentals of Statistical Signal Processing, Vol 1: Estimation Theory, Prentice Hall, Upper Saddle River

- It should be intuitively clear that because the value of θ affects the probability of $x[0]$, we should be able to *infer* the value of θ from the observed value of $x[0]$.
- For example, if the value of $x[0]$ is negative, it is doubtful that $\theta = \theta_2$. The value $\theta = \theta_1$ might be more reasonable.
- The specification of the PDF is critical in determining a good estimator.
- In an actual problem we are not given a PDF but must choose one that is not only consistent with the problem constraints, but one that is also mathematically tractable.

Estimation Theory – Classical Estimation Basics

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- Of course, the performance of any estimator obtained will be *critically dependent* on the PDF assumptions.
- We can only hope the estimator obtained is robust, in that slight changes in the PDF do not severely affect the performance of the estimator.
- Estimation based on PDFs like above is termed *classical* estimation in that the parameters of interest are assumed to be *deterministic* but unknown.
- In some applications we have a prior knowledge about the parameter to be estimated. E.g. for a given resistor R we might know that the mean is 1k and the variance has some value σ^2.
- To incorporate this prior knowledge, we can assume that R is no longer deterministic but a random variable and assign it a PDF, possibly $R \sim \mathcal{N}(1\text{k}, \sigma^2)$. Such an approach is termed *Bayesian* estimation.

Estimation Theory – Classical Estimation Basics

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- The parameter we are attempting to estimate is then viewed as a *realization* of the random variable.

- As such, the data are described by the *joint* PDF

$$p(\mathbf{x}, \theta) = p(\mathbf{x}|\theta)p(\theta)$$

  where $p(\theta)$ is the prior PDF, summarizing our knowledge about θ before any data are observed, and $p(\mathbf{x}|\theta)$ is a conditional PDF, summarizing our knowledge provided by the data $\mathbf{x}$ conditioned on knowing θ.

- The reader should compare the notational differences between $p(\mathbf{x}; \theta)$ (a family of PDFs) and $p(\mathbf{x}|\theta)$ (a conditional PDF).

Estimation Theory – Classical Estimation Basics

- Performance versus complexity:
  - Another theme that we will repeatedly encounter is the tradeoff between performance and computational complexity.
  - We will see that optimal estimators can sometimes be difficult to implement, requiring a multidimensional optimization or integration.
  - In these situations, alternative estimators that are suboptimal - but less complex - may be preferred.

Example: DC Level in Noise $x[n] = A + w[n]$, where $w[n]$ is zero mean uncorrelated noise with constant variance σ^2.

- **x** = [*0.9984, 0.9760, 1.0297, 1.0451, 0.9626, 0.9790, 1.0197, 0.9404, 1.0153, 0.9838*]^T
- Two possible candidates are

$$\hat{A}_1 = \frac{1}{N}\sum_{n=0}^{N-1} x[n] = 0.995 \qquad \hat{A}_2 = x[0] = 0.9984$$

Estimation Theory – Classical Estimation Basics

- Questions and Comments:
  - Which Estimator is better?
  - Intuitively we would not expect $\hat{A}_2$ to perform very well since there is no averaging of noise effects.
  - But for the presented example: $\hat{A}_2$ is closer to the true value of 1 than $\hat{A}_1$
  - So, is $\hat{A}_2$ better?
- ➔ *Need for a performance measure*
- One possibility: Evaluate performance with computer simulations
- To be preferred: Statistical measure of the estimation error
- First, we compare the means of the estimators:
  - $$E[\hat{A}_1] = E\left[\frac{1}{N}\sum_{n=0}^{N-1} x[n]\right] = \frac{1}{N}\sum_{n=0}^{N-1} E[x[n]] = \frac{1}{N}NA = A$$
  - $$E[\hat{A}_2] = E[x[0]] = A$$

Estimation Theory – Classical Estimation Basics – Unbiased Estimators

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- We say that both estimators are *unbiased* since they fulfill

$$E[\hat{\theta}] = \theta \qquad \text{for all } \theta.$$

- An unbiased estimator will on the average yield the true value of the unknown parameter.
- This restriction is already quite an important one. Letting $\hat{\theta} = g(\mathbf{x})$ it asserts that

$$E[\hat{\theta}] = \int_{-\infty}^{\infty} g(\mathbf{x}) p(\mathbf{x}; \theta) d\mathbf{x} = \theta \qquad \text{for all } \theta.$$

- In the next subsections we will see, that frequently we are looking for best possible unbiased estimators. This is often much easier than to look for the best possible biased estimators.

Estimation Theory – Classical Estimation Basics

- Next, we calculate the variances of the two estimators from above:

$$\mathrm{var}(\hat{A}_1) = \mathrm{var}\left(\frac{1}{N}\sum_{n=0}^{N-1} x[n]\right) = \frac{1}{N^2}\mathrm{var}\left(\sum_{n=0}^{N-1} x[n]\right)$$

$$= \frac{1}{N^2}\sum_{n=0}^{N-1}\mathrm{var}(x[n]) = \frac{1}{N^2}N\sigma^2 = \frac{\sigma^2}{N}$$

$$\mathrm{var}(\hat{A}_2) = \mathrm{var}(x[0]) = \sigma^2$$

- Hence $\hat{A}_1$ will on average produce estimates closer to the true value than $\hat{A}_2$.
- So, from a statistical performance point of view, $\hat{A}_1$ is to be preferred over $\hat{A}_2$.
- But the complexity of $\hat{A}_1$ is higher.

Estimation Theory – Classical Estimation Basics – Biased/Unbiased Estimators

- That an estimator is unbiased does not necessarily mean that it is a good estimator.
- It only guarantees that on the average it will attain the true value.
- On the other hand, biased estimators are ones that are characterized by a systematic error.
- The bias of an estimator is defined as

$$b(\theta) = E[\hat{\theta}] - \theta.$$

Estimation Theory – Classical Estimation Basics – The MSE Criterion

- In searching for optimal estimators, we need to adopt some optimality criterion.
- A natural one is the *mean square error* (MSE), defined as

$$\mathrm{mse}(\hat{\theta}) = E[(\hat{\theta} - \theta)^2].$$

- This measures the mean squared deviation of the estimator from the true value.
- Note that the MSE is in general a function of θ since

$$\mathrm{mse}(\hat{\theta}) = E[(\hat{\theta} - \theta)^2] = \int_{-\infty}^{\infty} (\hat{\theta} - \theta)^2 p(\mathbf{x}; \theta) d\mathbf{x}.$$

- Unfortunately, it is in general very difficult, and often impossible, to find the estimator that minimizes the MSE for all θ.

Estimation Theory – Classical Estimation Basics – The MSE Criterion

- Relationship between MSE, bias and variance:

$$\begin{aligned}\mathrm{mse}(\hat{\theta}) &= E[(\hat{\theta}-\theta)^2] \\ &= E[\{(\hat{\theta}-E[\hat{\theta}]) + \underbrace{(E[\hat{\theta}]-\theta)}_{b(\theta)}\}^2] \\ &= E[(\hat{\theta}-E[\hat{\theta}])^2 + 2(\hat{\theta}-E[\hat{\theta}])b(\theta) + b^2(\theta)] \\ &= \mathrm{var}(\hat{\theta}) + b^2(\theta)\end{aligned}$$

$$\boxed{\mathrm{mse}(\hat{\theta}) = \mathrm{var}(\hat{\theta}) + b^2(\theta)}$$

- The MSE is composed of errors due to the variance of the estimator as well as the bias.
- Interestingly, biased estimators might exist, that show better MSE performance than the best unbiased estimator (if it exists at all).

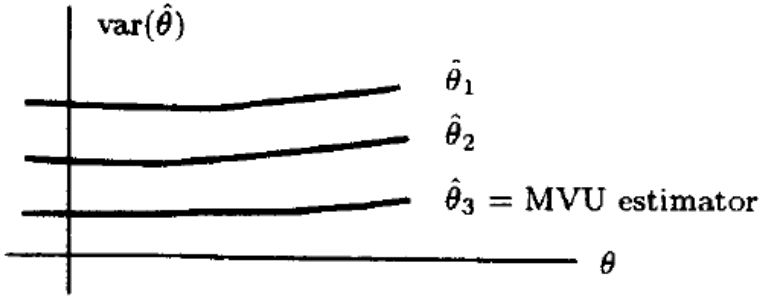
Estimation Theory – Classical Estimation
MVU Estimator

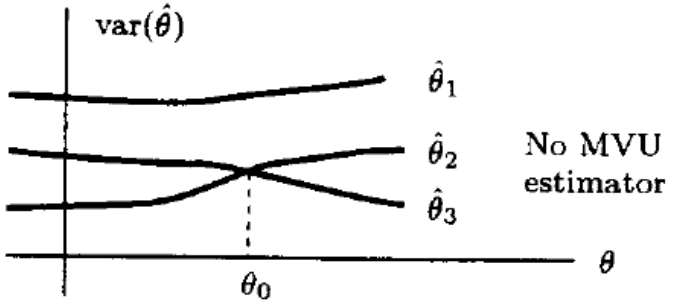
- Since it is often impossible to find the estimator that minimizes the MSE for all θ, an alternative approach is to constrain the bias to be zero and find the estimator which minimizes the variance.

- Such an estimator is termed the *minimum variance unbiased* (MVU) estimator.

- Note that the MSE of an unbiased estimator is just the variance.

- The question arises whether an MVU estimator exists, i.e., an unbiased estimator with minimum variance for all θ. Unfortunately, in general, the MVU estimator does not always exist. The following situations might appear:





Taken from [1].

Estimation Theory – Classical Estimation
MVU Estimator

- Two questions for a given problem at hand:
  - Does the MVU exist?
  - How to find it?

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Estimation Theory – Classical Estimation MVU estimator / CRLB

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Theorem (Cramer Rao Lower Bound – Scalar Parameter): *It is assumed that the PDF* $p(\mathbf{x};\theta)$ *satisfies the "regularity" condition* $E\left[\frac{\partial \ln p(\mathbf{x};\theta)}{\partial \theta}\right] = 0$ *for all* θ.

- *Then the variance of any unbiased estimator* $\hat{\theta}$ *satisfies*

$$\mathrm{var}(\hat{\theta}) \geq \frac{1}{-E\left[\frac{\partial^2 \ln p(\mathbf{x};\theta)}{\partial \theta^2}\right]} = \frac{1}{I(\theta)},$$

where the derivative is evaluated at the true value of θ *and the expectation is taken with respect to* $p(\mathbf{x};\theta)$. $I(\theta) = -E\left[\frac{\partial^2 \ln p(\mathbf{x};\theta)}{\partial \theta^2}\right]$ *is called Fisher information.*

- *Furthermore, an unbiased estimator may be found that attains the CRLB for all* θ *if and only if*

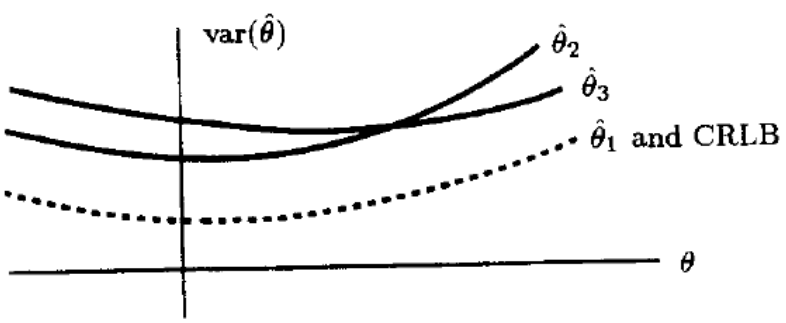
$$\frac{\partial \ln p(\mathbf{x};\theta)}{\partial \theta} = I(\theta)(g(\mathbf{x}) - \theta)$$

for some functions g *and* I. *That estimator, which is the MVU estimator, is* $\hat{\theta} = \mathrm{g}(\mathbf{x})$, *and the minimum variance is* $1/I(\theta)$.

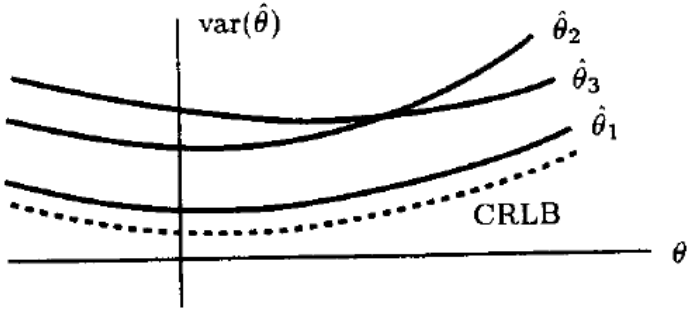
- The derivation of the CRLB theorem can be found in the appendix.
- We will extend the CRLB statements to the case of a vector parameter immediately.
- A number of consequences to the scalar and vector CRLB theorems can be found in [1], some of them not covered in the lecture will be discussed in the exercises.

Estimation Theory – Classical Estimation
MVU estimator / CRLB

- An unbiased estimator is said to be *efficient* if it attains the CRLB.
- An efficient estimator is always the MVU estimator, but the MVU is not necessarily an efficient estimator!
- The following situations might appear:



(a) $\hat{\theta}_1$ efficient and MVU



(b) $\hat{\theta}_1$ MVU but not efficient

Taken from [1].

Discussion of the regularity condition

$$E\left[\frac{\partial \ln p(\mathbf{x};\theta)}{\partial \theta}\right] = \int \frac{\partial \ln p(\mathbf{x};\theta)}{\partial \theta} p(\mathbf{x};\theta)\mathrm{d}\mathbf{x}$$

$$= \int \frac{1}{p(\mathbf{x};\theta)} \frac{\partial p(\mathbf{x};\theta)}{\partial \theta} p(\mathbf{x};\theta)\mathrm{d}\mathbf{x} \qquad \text{(chain rule)}$$

$$= \int \frac{\partial p(\mathbf{x};\theta)}{\partial \theta}\mathrm{d}\mathbf{x}$$

- If the order of the integration and the differentiation may be interchanged we obtain

$$E\left[\frac{\partial \ln p(\mathbf{x};\theta)}{\partial \theta}\right] = \frac{\partial}{\partial \theta}\int p(\mathbf{x};\theta)\mathrm{d}\mathbf{x} = \frac{\partial 1}{\partial \theta} = 0$$

Estimation Theory – Classical Estimation
MVU estimator / CRLB

- We can conclude that the regularity condition will be satisfied if the order of the integration and the differentiation may be interchanged.
- This is generally true except when the domain for which the PDF is nonzero depends on θ (cf. Leibniz integral rule).

Derivation of the CRLB

- We consider all unbiased estimators, or those for which

$$E\left[\hat{\theta}\right] = \theta$$

$$\int \hat{\theta} p(\mathbf{x};\theta)\mathrm{d}\mathbf{x} = \theta$$

- Now we differentiate both sides with respect to θ:

$$\frac{\partial}{\partial\theta}\int \hat{\theta} p(\mathbf{x};\theta)\mathrm{d}\mathbf{x} = 1$$

- Interchanging the partial differentiation and integration produces

$$\int \hat{\theta}\frac{\partial p(\mathbf{x};\theta)}{\partial\theta}\mathrm{d}\mathbf{x} = 1$$

- or

$$\int \hat{\theta} \frac{\partial \ln p(\mathbf{x};\theta)}{\partial \theta} p(\mathbf{x};\theta) \mathrm{d}\mathbf{x} = 1$$

Note: $\frac{\partial \ln p(\mathbf{x};\theta)}{\partial \theta} = \frac{1}{p(\mathbf{x};\theta)} \frac{\partial p(\mathbf{x};\theta)}{\partial \theta}$

- We now use the regularity condition

$$E\left[\frac{\partial \ln p(\mathbf{x};\theta)}{\partial \theta}\right] = \int \frac{\partial \ln p(\mathbf{x};\theta)}{\partial \theta} p(\mathbf{x};\theta) \mathrm{d}\mathbf{x} = 0$$

- multiply it with θ to obtain

$$\theta \int \frac{\partial \ln p(\mathbf{x};\theta)}{\partial \theta} p(\mathbf{x};\theta) \mathrm{d}\mathbf{x} = \int \theta \frac{\partial \ln p(\mathbf{x};\theta)}{\partial \theta} p(\mathbf{x};\theta) \mathrm{d}\mathbf{x} = 0$$

- and modify the equation at the top to produce

$$\int (\hat{\theta} - \theta) \frac{\partial \ln p(\mathbf{x};\theta)}{\partial \theta} p(\mathbf{x};\theta) \mathrm{d}\mathbf{x} = 1$$

- We now apply a version of the Cauchy-Schwarz inequality

$$\left(\int w(\mathbf{x})g(\mathbf{x})h(\mathbf{x})\mathrm{d}\mathbf{x}\right)^2 \leq \left(\int w(\mathbf{x})g^2(\mathbf{x})\mathrm{d}\mathbf{x}\right)\cdot\left(\int w(\mathbf{x})h^2(\mathbf{x})\mathrm{d}\mathbf{x}\right)$$

  which holds with equality if and only if $g(\mathbf{x}) = c \cdot h(\mathbf{x})$ for c some constant not depending on $\mathbf{x}$. The functions g and h are arbitrary functions, while $w(\mathbf{x})$ needs to be non-negative for all $\mathbf{x}$.

- Now let $w(\mathbf{x}) = p(\mathbf{x};\theta)$, $g(\mathbf{x}) = \hat{\theta} - \theta$, $h(\mathbf{x}) = \frac{\partial \ln p(\mathbf{x};\theta)}{\partial\theta}$ and apply the Cauchy-Schwarz inequality:

$$1^2 = \left(\int (\hat{\theta}-\theta)\frac{\partial \ln p(\mathbf{x};\theta)}{\partial\theta}p(\mathbf{x};\theta)\mathrm{d}\mathbf{x}\right)^2$$

$$\leq \left(\int (\hat{\theta}-\theta)^2 p(\mathbf{x};\theta)\mathrm{d}\mathbf{x}\right)\cdot\left(\int \left(\frac{\partial \ln p(\mathbf{x};\theta)}{\partial\theta}\right)^2 p(\mathbf{x};\theta)\mathrm{d}\mathbf{x}\right)$$

- or

$$\mathrm{var}(\hat{\theta}) \geq \frac{1}{E\left[\left(\frac{\partial \ln p(\mathbf{x};\theta)}{\partial \theta}\right)^2\right]}$$

- In the following we will show that

$$E\left[\left(\frac{\partial \ln p(\mathbf{x};\theta)}{\partial \theta}\right)^2\right] = -E\left[\frac{\partial^2 \ln p(\mathbf{x};\theta)}{\partial \theta^2}\right]$$

- We start with the regularity condition:

$$E\left[\frac{\partial \ln p(\mathbf{x};\theta)}{\partial \theta}\right] = 0$$

$$\int \frac{\partial \ln p(\mathbf{x};\theta)}{\partial \theta} p(\mathbf{x};\theta) \mathrm{d}\mathbf{x} = 0$$

$$\frac{\partial}{\partial\theta}\int \frac{\partial \ln p(\mathbf{x};\theta)}{\partial\theta} p(\mathbf{x};\theta)\mathrm{d}\mathbf{x} = 0$$

$$\int \left[\frac{\partial^2 \ln p(\mathbf{x};\theta)}{\partial\theta^2} p(\mathbf{x};\theta) + \frac{\partial \ln p(\mathbf{x};\theta)}{\partial\theta}\frac{\partial p(\mathbf{x};\theta)}{\partial\theta}\right]\mathrm{d}\mathbf{x} = 0$$

$$\int \left[\frac{\partial^2 \ln p(\mathbf{x};\theta)}{\partial\theta^2} p(\mathbf{x};\theta) + \left(\frac{\partial \ln p(\mathbf{x};\theta)}{\partial\theta}\right)^2 p(\mathbf{x};\theta)\right]\mathrm{d}\mathbf{x} = 0$$

$$E\left[\left(\frac{\partial \ln p(\mathbf{x};\theta)}{\partial\theta}\right)^2\right] = -E\left[\frac{\partial^2 \ln p(\mathbf{x};\theta)}{\partial\theta^2}\right]$$

- Thus

$$\mathrm{var}(\hat{\theta}) \geq \frac{1}{-E\left[\frac{\partial^2 \ln p(\mathbf{x};\theta)}{\partial\theta^2}\right]}$$

- Note that the condition for equality in the Cauchy-Schwarz inequality is

$$\frac{\partial \ln p(\mathbf{x};\theta)}{\partial \theta} = \frac{1}{c}(\hat{\theta} - \theta)$$

- Here c cannot depend on $\mathbf{x}$, but it may depend on θ. We therefore write $c(\theta)$. To determine $c(\theta)$ we differentiate with respect to θ

$$\frac{\partial^2 \ln p(\mathbf{x};\theta)}{\partial \theta^2} = -\frac{1}{c(\theta)} + \frac{\partial\left(\frac{1}{c(\theta)}\right)}{\partial \theta}(\hat{\theta} - \theta)$$

- and take the expected value to get

$$E\left[\frac{\partial^2 \ln p(\mathbf{x};\theta)}{\partial \theta^2}\right] = -\frac{1}{c(\theta)} \quad \text{or} \quad c(\theta) = \frac{1}{-E\left[\frac{\partial^2 \ln p(\mathbf{x};\theta)}{\partial \theta^2}\right]} = \frac{1}{I(\theta)}$$

■ **Example: DC Level in white Gaussian noise (WGN)**

Let $x[n] = A + w[n]$ for $n = 0, \ldots, N-1$, where $w[n]$ is WGN with variance σ^2. Find the CRLB as well as the MVU estimator for A.

□ Each $x[n]$ is Gaussian distributed with

$$p(x[n]; A) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(x[n] - A)^2\right).$$

□ Since the noise samples are uncorrelated, those of $x[n]$ are also uncorrelated and (because of the Gaussian distribution) statistically independent. The joint PDF $p(\mathbf{x}; A)$ therefore is

$$p(\mathbf{x}; A) = \frac{1}{(2\pi\sigma^2)^{\frac{N}{2}}} \exp\left(-\frac{1}{2\sigma^2}\sum_{n=0}^{N-1}(x[n] - A)^2\right)$$

- Taking the first derivative of $\ln p\,(\mathbf{x}; A)$ yields

$$\frac{\partial \ln p\,(\mathbf{x}; A)}{\partial A} = \frac{\partial}{\partial A}\left(-\ln[\,(2\pi\sigma^2)^{\frac{N}{2}}] - \frac{1}{2\sigma^2}\sum_{n=0}^{N-1}(x[n]-A)^2 \right) = \frac{1}{\sigma^2}\sum_{n=0}^{N-1}(x[n]-A)$$

$$= \frac{N}{\sigma^2}\left(\frac{1}{N}\sum_{n=0}^{N-1} x[n] - A \right) = \frac{N}{\sigma^2}(\bar{x} - A)$$

with the *sample mean*

$$g(\mathbf{x}) = \bar{x} = \frac{1}{N}\sum_{n=0}^{N-1} x[n]\,.$$

- Differentiating again gives

$$\frac{\partial^2 \ln p\,(\mathbf{x}; A)}{\partial A^2} = -\frac{N}{\sigma^2}.$$

Estimation Theory – Classical Estimation MVU estimator / CRLB

JKU

- This leads to the CRLB

$$\mathrm{var}(\hat{A}) \geq \frac{\sigma^2}{N}.$$

- Let's again look at the result from the first derivative:

$$\frac{\partial \ln p(\mathbf{x}; A)}{\partial A} = \frac{N}{\sigma^2}(\bar{x} - A) = I(A)(g(\mathbf{x}) - A)$$

- With the second part of the CRLB theorem we can conclude that for a DC level in WGN the sample mean

$$\hat{A} = g(\mathbf{x}) = \bar{x} = \frac{1}{N}\sum_{n=0}^{N-1} x[n]$$

is the MVU estimator with variance $\mathrm{var}(\hat{A}) = \frac{\sigma^2}{N}$.

Example: Phase Estimation

Let

$$x[n] = A\cos(2\pi f_0 n + \varphi) + w[n]$$

for $n = 0, \ldots, N-1$, with known amplitude A, known (normalized) frequency f_0, and WGN $w[n]$ with variance σ^2. Find the CRLB for the phase φ.

Solution: It can be shown that for f_0 not near to 0 or 1/2 the CRLB is approximately

$$\mathrm{var}(\hat{\varphi}) \geq \frac{2\sigma^2}{NA^2}.$$

(Note that during the derivation you should use the approximation $N^{-1}\sum_{n=0}^{N-1}\cos(4\pi f_0 n + 2\varphi) \approx 0$). We furthermore note (without proof), that in this case no efficient estimator exists.

Example: Frequency Estimation

Let

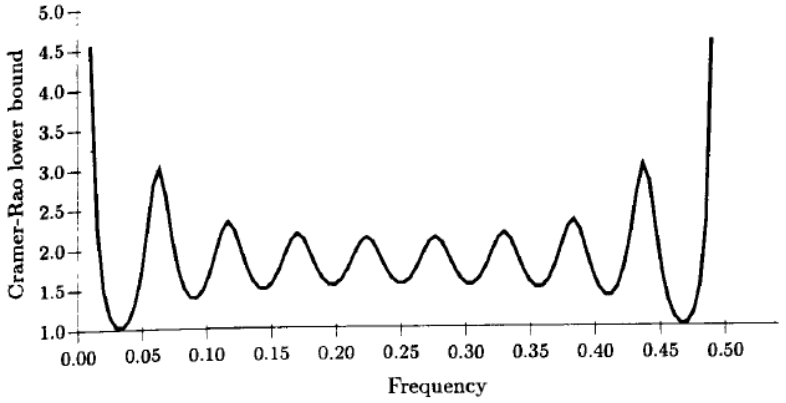
$$x[n] = A\cos(2\pi f_0 n + \varphi) + w[n] \qquad 0 < f_0 < 1/2$$

for $n = 0, \ldots, N-1$, with known amplitude A, known phase φ, and WGN $w[n]$ with variance σ^2. Find the CRLB for the normalized frequency f_0.

Solution: It can be shown that

$$\mathrm{var}(\hat{f}_0) \geq \frac{\sigma^2}{A^2 \sum_{n=0}^{N-1} [2\pi n \sin(2\pi f_0 n + \varphi)]^2}.$$

This is an example, where the CRLB depends on the true value of the parameter to be estimated.



Taken from [1].

Transformation of parameters

Assume we wish to estimate a parameter α that is a function $g(\theta)$ of some more fundamental parameter θ, and we already know the CRLB for θ.

- Then the CRLB for $g(\theta)$ can be obtained by (without proof):

$$\mathrm{var}(\hat{\alpha}) \geq \frac{\left(\frac{\partial g}{\partial \theta}\right)^2}{-E\left[\frac{\partial^2 \ln p(\mathbf{x};\theta)}{\partial \theta^2}\right]}$$

- **Example**: If we wish to estimate $\alpha = g(A) = A^2$ in the DC in WGN example, then the CRLB for α becomes

$$\mathrm{var}(\hat{\alpha}) \geq \frac{(2A)^2}{N/\sigma^2} = \frac{4A^2\sigma^2}{N}.$$

Transformation of parameters

- Note that efficiency is in general destroyed by non-linear transformations.
  - The sample mean estimator $\bar{x}$ was efficient for A.
  - We now regard the estimator $\hat{\alpha} = \bar{x}^2$ for A^2.
  - This estimator is not even unbiased since

$$E[\hat{\alpha}] = E[\bar{x}^2] = E^2[\bar{x}] + \mathrm{var}(\bar{x}) = A^2 + \frac{\sigma^2}{N} \neq A^2.$$

- It can easily be shown that efficiency is preserved by linear (actually affine) transformations of the form $\alpha = a\theta + b$, i.e., if $\hat{\theta}$ is an efficient estimator for θ, then $\hat{\alpha} = a\hat{\theta} + b$ is an efficient estimator for $\alpha = a\theta + b$.

Estimation Theory – Classical Estimation
MVU estimator / CRLB for Vector Parameter

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- We now extend the results of the previous sections to the case where we wish to estimate a vector parameter $\boldsymbol{\theta} = [\theta_1, \theta_2, \ldots, \theta_p]^T$.

- We again look for unbiased estimators. We say that an estimator for a vector parameter is unbiased if

$$E[\hat{\theta}_i] = \theta_i \quad \text{for all } \theta_i.$$

- This is equivalent to

$$E[\widehat{\boldsymbol{\theta}}] = \boldsymbol{\theta} \quad \text{for all } \boldsymbol{\theta}.$$

- The MVU estimator has the additional property that $\mathrm{var}(\hat{\theta}_i)$ for $i = 1,2,\ldots,p$ is minimum among all unbiased estimators.

Estimation Theory – Classical Estimation
MVU estimator / CRLB for Vector Parameter

Theorem (Cramer Rao Lower Bound – Vector Parameter): *It is assumed that the PDF* $p(\mathbf{x};\boldsymbol{\theta})$ *satisfies the "regularity" condition* $E\left[\frac{\partial \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\right] = \mathbf{0}$ *for all* $\boldsymbol{\theta}$.

- *Then the covariance matrix of any unbiased estimator* $\hat{\boldsymbol{\theta}}$ *satisfies*

$$\mathbf{C}_{\hat{\boldsymbol{\theta}}\hat{\boldsymbol{\theta}}} - \mathbf{I}^{-1}(\boldsymbol{\theta}) \geq \mathbf{0}$$

where $\geq \mathbf{0}$ *is interpreted as meaning that the matrix* is *positive semi-definite. The so called Fisher information matrix* $\mathbf{I}(\boldsymbol{\theta})$ *is given as* $[\mathbf{I}(\boldsymbol{\theta})]_{ij} = -E\left[\frac{\partial^2 \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial \theta_i \partial \theta_j}\right]$, *where the derivatives are evaluated at the true value of* $\boldsymbol{\theta}$, *and the expectation is taken with respect* to $p(\mathbf{x};\boldsymbol{\theta})$.

- *Furthermore, an unbiased estimator may be found that attains the CRLB for all* $\boldsymbol{\theta}$ *in that* $\mathbf{C}_{\hat{\boldsymbol{\theta}}\hat{\boldsymbol{\theta}}} = \mathbf{I}^{-1}(\boldsymbol{\theta})$ *if and only if*

$$\frac{\partial \ln p(\mathbf{x};\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = \mathbf{I}(\boldsymbol{\theta})(\mathbf{g}(\mathbf{x}) - \boldsymbol{\theta})$$

for some p-dimensional function $\mathbf{g}$ *and some p x p matrix* $\mathbf{I}$. *That estimator, which is the MVU estimator, is* $\hat{\boldsymbol{\theta}} = \mathbf{g}(\mathbf{x})$, *and its covariance matrix is* $\mathbf{C}_{\hat{\boldsymbol{\theta}}\hat{\boldsymbol{\theta}}} = \mathbf{I}^{-1}(\boldsymbol{\theta})$.

- An important consequence of

$$\mathbf{C}_{\hat{\boldsymbol{\theta}}\hat{\boldsymbol{\theta}}} - \mathbf{I}^{-1}(\boldsymbol{\theta}) \geq \mathbf{0}$$

is, that we can place a bound on the variance of each element of the parameter vector.

- This follows since for a positive semi-definite matrix the diagonal elements are nonnegative. Hence

$$\left[\mathbf{C}_{\hat{\boldsymbol{\theta}}\hat{\boldsymbol{\theta}}} - \mathbf{I}^{-1}(\boldsymbol{\theta})\right]_{ii} \geq 0$$

and therefore

$$\mathrm{var}(\hat{\theta}_i) = \left[\mathbf{C}_{\hat{\boldsymbol{\theta}}\hat{\boldsymbol{\theta}}}\right]_{ii} \geq \left[\mathbf{I}^{-1}(\boldsymbol{\theta})\right]_{ii}.$$

Problem: Sinusoidal Parameter Estimation

Let

$$x[n] = A\cos(2\pi f_0 n + \phi) + w[n] \qquad 0 < f_0 < 1/2$$

for $n = 0, \ldots, N-1$, with A, φ, f_0 unknown, and WGN $w[n]$ with variance σ^2. Find the CRLBs for A, φ and f_0.

Solution: Using some approximations, and assuming that f_0 is not near to 0 or 1/2 the CRLBs can be found to be

$$\mathrm{var}(\hat{A}) \geq \frac{2\sigma^2}{N},$$

$$\mathrm{var}(\hat{f}_0) \geq \frac{24\sigma^2}{(2\pi)^2 A^2 N(N^2-1)}$$

$$\mathrm{var}(\hat{\phi}) \geq \frac{4\sigma^2(2N-1)}{A^2 N(N+1)}.$$